

Iris Dataset

Exploratory Data Analysis Report

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Dataset: 150 observations $\times$ 5 features

1. Introduction

The Iris dataset is one of the most widely used datasets in machine learning and statistics. It was introduced by the British statistician and biologist Ronald Fisher in 1936. The dataset contains measurements of four features from 150 iris flowers belonging to three different species.

1.1 Dataset Overview

Total Observations: 150

Total Features: 5 (4 numeric + 1 target)

Species: 3 (Setosa, Versicolor, Virginica)

Samples per Species: 50 (perfectly balanced)

Missing Values: None

File Size: 6.0 KB

1.2 Features

The dataset includes the following four numerical features measured in centimetres:

- Sepal Length (cm) — length of the sepal
- Sepal Width (cm) — width of the sepal
- Petal Length (cm) — length of the petal
- Petal Width (cm) — width of the petal
- Species — encoded as 0 (Setosa), 1 (Versicolor), 2 (Virginica)

2. Statistical Summary

The table below presents descriptive statistics for all numerical features across the full dataset.

Statistic	Sepal Len	Sepal Wid	Petal Len	Petal Wid
Count	150	150	150	150
Mean	5.843	3.057	3.758	1.199
Std Dev	0.828	0.436	1.765	0.762
Min	4.300	2.000	1.000	0.100
25%	5.100	2.800	1.600	0.300
Median	5.800	3.000	4.350	1.300
75%	6.400	3.300	5.100	1.800
Max	7.900	4.400	6.900	2.500

Table 1: Descriptive statistics for all numeric features

Key observations from the statistical summary:

- Petal Length has the highest variability (std = 1.765), making it the most discriminating feature.
- Sepal Width has the lowest variability (std = 0.436), indicating it is more consistent across species.
- Petal Length and Petal Width show large ranges (1.0–6.9 cm and 0.1–2.5 cm respectively), suggesting strong between-species differences.
- Sepal Length is moderately variable (std = 0.828), with values ranging from 4.3 to 7.9 cm.

3. Data Visualisations

3.1 Species Distribution

The bar chart below confirms the perfectly balanced dataset: each of the three species (Setosa=0, Versicolor=1, Virginica=2) contains exactly 50 observations. This balance eliminates class-imbalance concerns in subsequent modelling tasks.

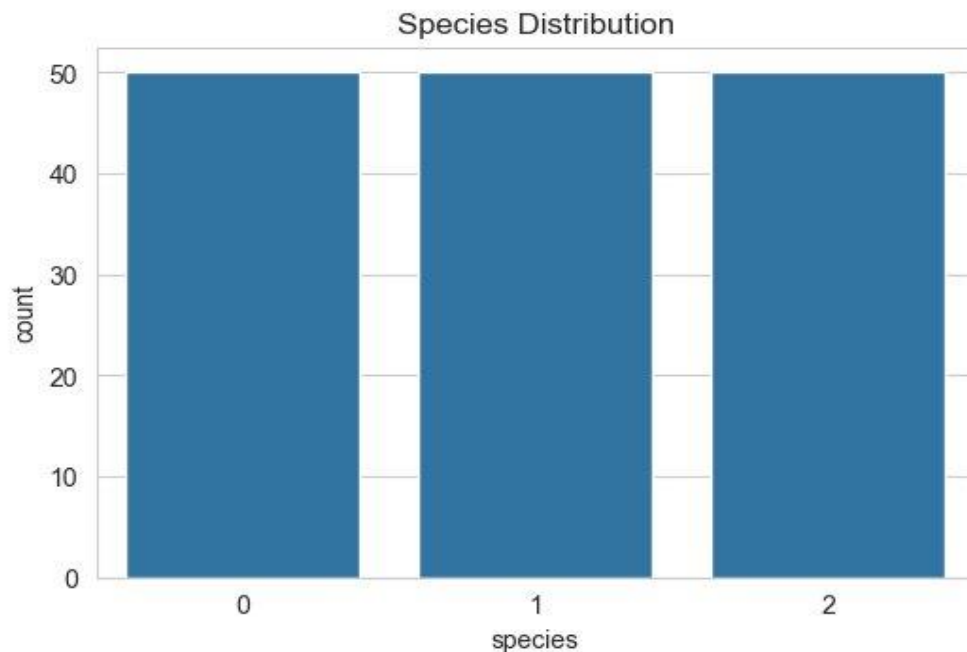


Figure 1: Balanced class distribution — 50 samples per species

3.2 Feature Distributions (Histograms)

Histograms reveal the marginal distribution of each feature. Petal Length and Petal Width exhibit a clear bimodal pattern, corresponding to Setosa (small petals) versus the other two species (larger petals). Sepal Length is approximately normally distributed, while Sepal Width is slightly right-skewed.

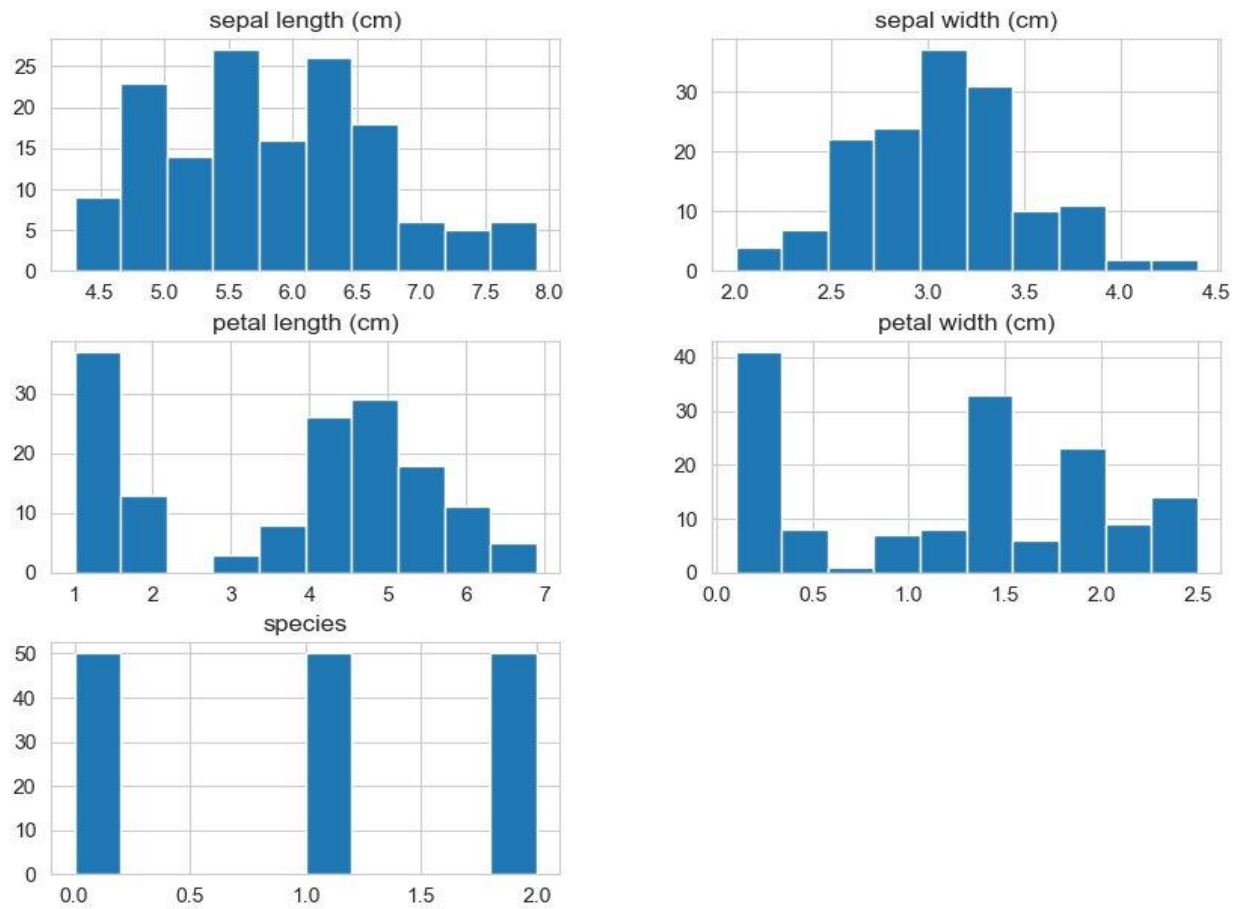


Figure 2: Histograms for all features — note bimodal distributions in petal dimensions

3.3 Feature Distribution & Outliers (Boxplots)

Boxplots provide a compact summary of central tendency, spread, and outliers for each feature. Petal Length shows the widest IQR (1.6–5.1 cm), confirming its high variance. Sepal Width has the tightest IQR and contains two mild outliers around 4.1–4.4 cm. Overall outlier prevalence is low, indicating clean data quality.

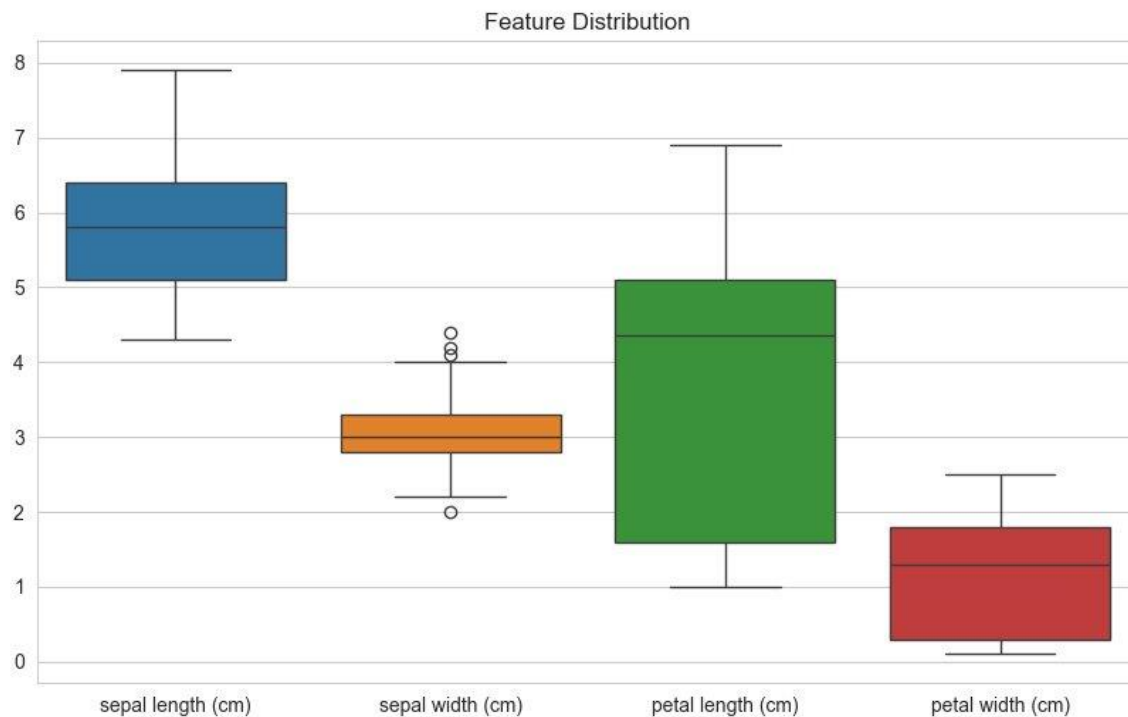


Figure 3: Boxplots showing feature spread — Petal Length has the largest IQR

3.4 Correlation Heatmap

The heatmap below displays Pearson correlation coefficients between all feature pairs. Strong positive correlations (dark red) exist between petal dimensions, while sepal width shows slight negative correlations with the other three features.

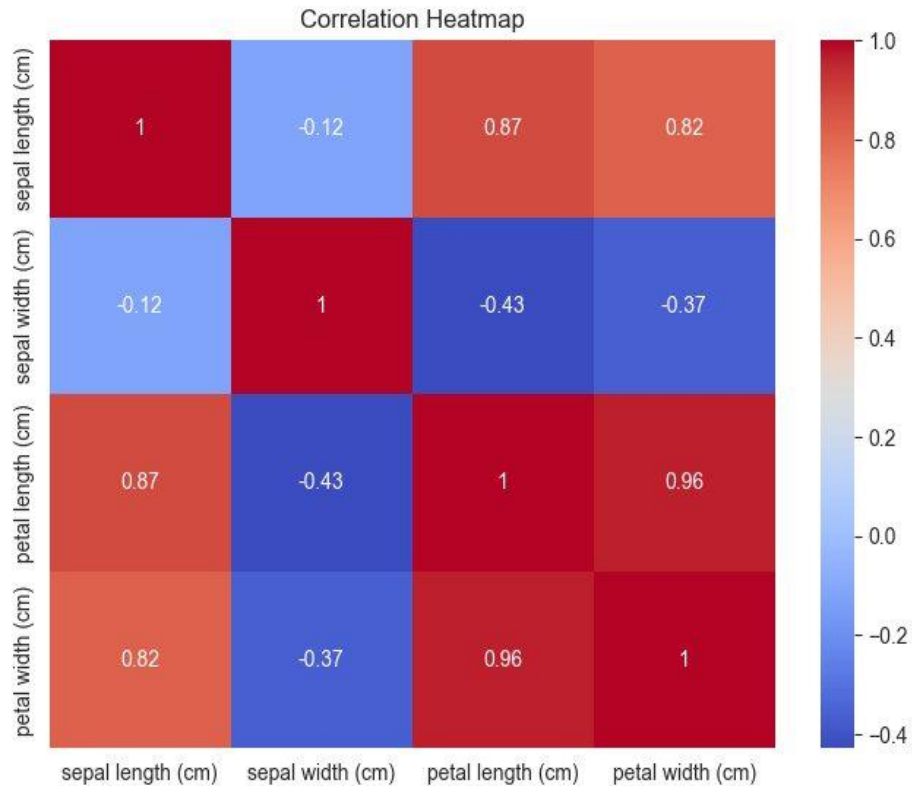


Figure 4: Pearson correlation heatmap

Notable correlation findings:

- Petal Length ↔ Petal Width: $r = 0.96$ (very strong positive) — these two features scale together.
- Sepal Length ↔ Petal Length: $r = 0.87$ (strong positive) — larger flowers tend to have longer sepals and petals.
- Sepal Length ↔ Petal Width: $r = 0.82$ (strong positive).
- Sepal Width ↔ Petal Length: $r = -0.43$ (moderate negative) — wider sepals occur in smaller-petalled flowers (Setosa).
- Sepal Width ↔ Sepal Length: $r = -0.12$ (near-zero) — sepal dimensions are largely independent.

3.5 Pairplot (All Feature Combinations)

The pairplot matrix shows scatter plots for every combination of feature pairs, with KDE curves on the diagonal. Species are colour-coded. Setosa (light pink) is clearly linearly separable from Versicolor (purple) and Virginica (dark) in petal-space, while the latter two overlap modestly.

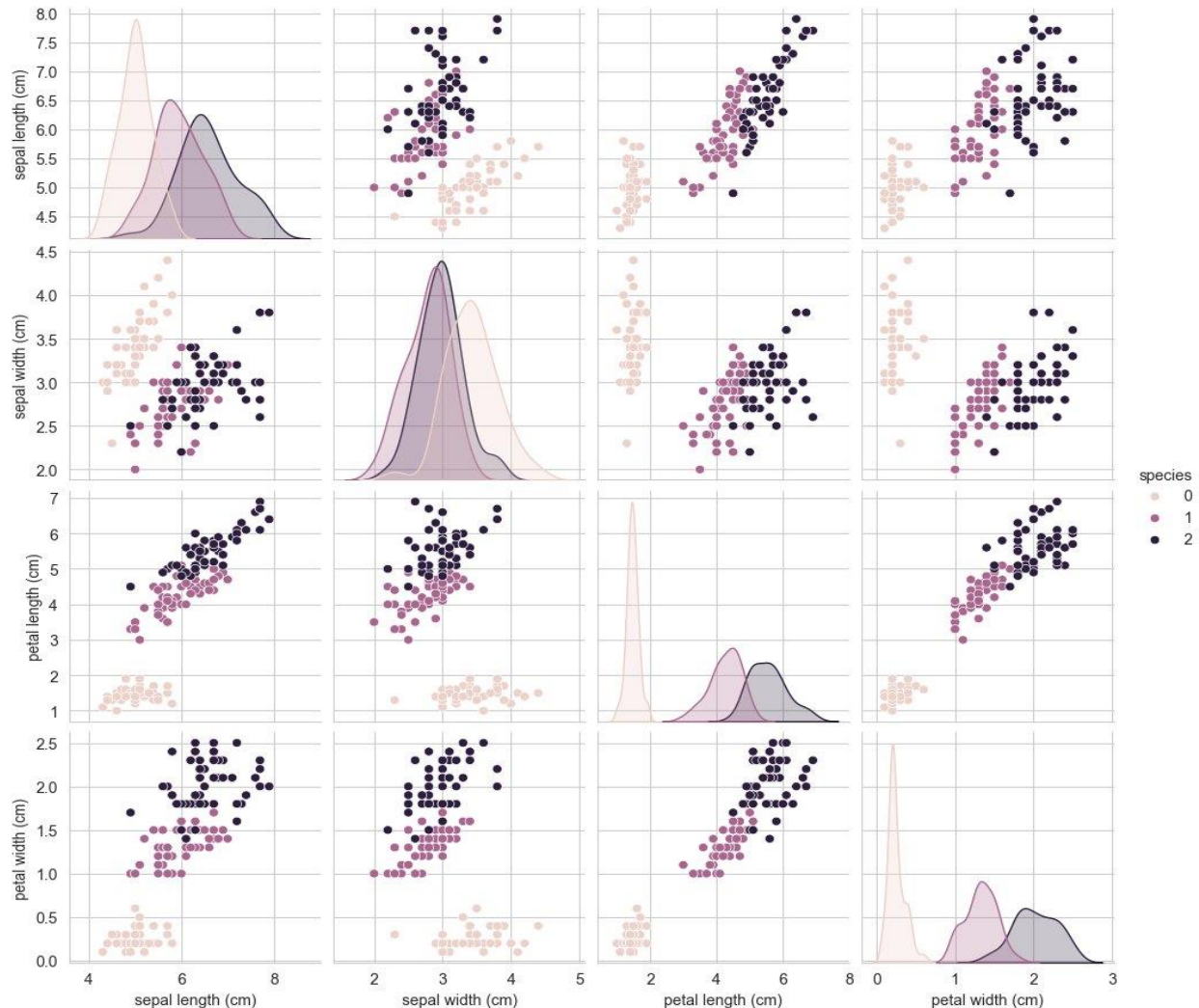


Figure 5: Pairplot matrix — Setosa is clearly separable in all petal-related axes

3.6 Sepal Length vs Petal Length (Scatter)

The scatter plot of Sepal Length vs Petal Length demonstrates strong species clustering. Setosa occupies the bottom-left (small petal, small sepal), Virginica the top-right (large petal, large sepal), and Versicolor lies in between. This confirms these two features alone can achieve strong species discrimination.

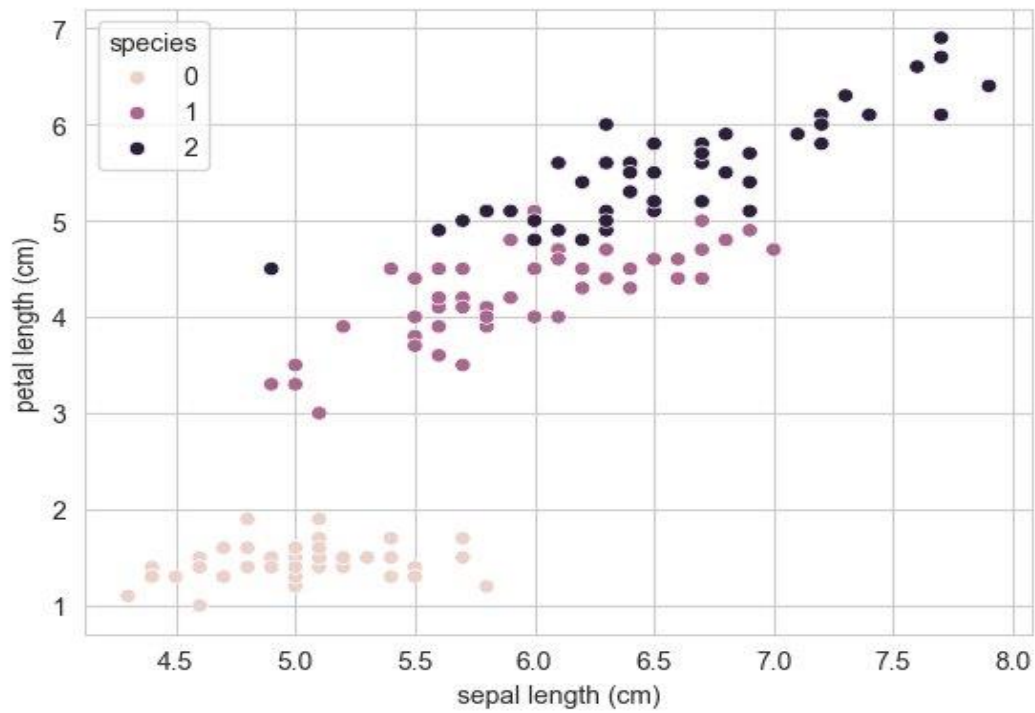


Figure 6: Sepal Length vs Petal Length — clear cluster separation across species

4. Key Findings & Insights

The following key findings emerged from the exploratory analysis:

1. Perfectly Balanced Dataset

Each species contains exactly 50 samples (150 total), ensuring unbiased model training with no class-imbalance correction required.

2. No Missing Values

All 150 rows across all 5 columns are complete, eliminating the need for imputation strategies.

3. Setosa is Linearly Separable

Setosa can be separated from Versicolor and Virginica using petal dimensions alone. This makes it trivially classifiable by simple linear models.

4. Petal Features are Most Discriminating

Petal Length and Petal Width show the highest variance and the strongest inter-species differences, making them the most valuable features for classification.

5. Strong Petal Correlation ($r = 0.96$)

Petal Length and Petal Width are almost perfectly correlated. In dimensionality reduction (e.g. PCA), they would likely combine into a single component.

6. Versicolor–Virginica Overlap

These two species overlap in sepal space, but are distinguishable in petal-length/width space. Non-linear models may be needed for perfect separation.

7. Virginica has Largest Dimensions

On average, Virginica has the longest and widest petals and sepals, making it the 'largest' species in the dataset.

5. Recommendations for Modelling

Based on the EDA findings, the following recommendations are made for downstream modelling:

- **Feature Selection:** Prioritise Petal Length and Petal Width as primary features. They offer the strongest class discrimination and account for most variance.
- **Dimensionality Reduction:** Consider applying PCA to reduce the four features to 2–3 principal components, given the high petal-feature correlation ($r = 0.96$).
- **Algorithm Choice:** Simple classifiers (Logistic Regression, Linear SVM) will likely achieve high accuracy. For Versicolor–Virginica separation, try k-NN, Decision Trees, or RBF-SVM.
- **No Scaling Required for Trees:** Tree-based models (Decision Tree, Random Forest) are scale-invariant. For distance-based models (k-NN, SVM), standardise features using z-score normalisation.
- **Baseline Target:** Expect classification accuracy above 95% given the strong cluster separation observed. Setosa should achieve 100% precision/recall.
- **Cross-Validation:** Use stratified k-fold CV ($k=5$ or $k=10$) to maintain class proportions in each fold.

6. Conclusion

The Iris dataset is a clean, balanced, and well-structured dataset ideal for classification tasks. The EDA reveals that petal dimensions are the primary discriminators between species, with Setosa being clearly separable and Versicolor/Virginica overlapping in sepal space. The strong correlation between petal features ($r = 0.96$) suggests potential for effective dimensionality reduction.

The absence of missing values and outliers (beyond 2 mild sepal-width outliers) means the data is ready for modelling with minimal preprocessing. Classification models leveraging petal features are expected to achieve accuracy well above 95%.